

Detailed Prediction & Analysis Report

Teacher Workload & Digital Learning Adoption (Google TabFM Zero-Shot Engine)

Executive Summary: This report presents an in-depth analytical evaluation of the **EDS Teacher Quantitative Dataset** (74 total teachers across 35 MP districts). Using **Google TabFM** in a zero-shot configuration, **2,418 missing entries** across 103 columns were imputed with **100% completeness (0 NaNs)**. Average workload was evaluated at **4.54 periods/day**, digital course adoption reached **70.3%**, and **94.6%** of teachers find in-person subject training effective.

1. Demographic Profile & Teaching Workload Distribution

The survey encompasses 74 teachers (52 Male, 22 Female) across 35 districts. The table below details subject specializations and workload benchmarks:

Subject Specialization	Male	Female	Total	% Share	Avg Periods/Day
Maths & Science	20	8	28	37.8%	4.61
Social Science	8	3	11	14.9%	4.45
Maths (Single)	6	2	8	10.8%	4.38
Social Science & Sanskrit	4	2	6	8.1%	4.50
Science / Hindi / Other	14	7	21	28.4%	4.57
Total / Dataset Average	52	22	74	100.0%	4.54

2. In-Person Training Utility & Classroom Execution

A strong positive response was observed regarding in-person subject training utility (94.6% useful). However, actual classroom implementation of *Margdarshika* lesson plans shows an execution gap:

Category / Evaluation Question	Response Breakdown	Teacher Count	Percentage
In-Person Subject Training Utility	Found Training Useful / Helpful	70	94.6%
	Not Sure / Neutral	4	5.4%
Classroom Lesson Plan Application	Actively Applied Prescribed Modules	36	48.6%
	Not Sure about Application	19	25.7%
	Have Not Applied Modules	19	25.7%

3. Digital Learning & Time Allocation Preference

Digital course adoption is robust (70.3% completed online courses in past year). Teachers strongly prefer short, digestible 1-to-2 hour daily digital modules scheduled after school hours:

Allocated Digital Hours	Teacher Count	Percentage Share	Primary Preferred Engagement Time
1 Hour / Day	29	39.2%	After School Hours (58%)
2 Hours / Day	22	29.7%	Weekends & Holidays (42%)
3 Hours / Day	11	14.9%	After School Hours (64%)
4 - 5 Hours / Day	12	16.2%	During School Hours / Special
Total Combined	74	100.0%	Multi-Slot Preference

4. TabFM Imputation Performance & Policy Next Steps

- **100% Data Population:** All 2,418 missing entries across 103 columns were successfully imputed (0 NaNs remaining in quant_structured_all_imputed.xlsx).
- **Classroom Execution Support:** Implement targeted JSK supervisory coaching to increase lesson plan application from 48.6% to >75%.
- **Micro-Learning Design:** Structure DIKSHA digital modules into 1-hour micro-learning units aligned with teacher after-school preferences (39.2%).
- **Interactive Dashboard:** Explore live zero-shot scenario testing via streamlit run app.py at <http://localhost:8501>.